Multi Tier Computing

Class Tutorial – 1

Question 1: Explore Myths and Challenges of Client/Server Computing with Justification

Client/Server computing is often misunderstood due to oversimplified views presented in marketing literature and incomplete interpretations.

Myths of Client/Server Computing

- **Client/Server computing is easy to implement:** In reality, integrating hardware and software from multiple vendors is complex and requires careful planning and expertise across platforms.
- **Existing desktop machines are sufficient:** Older systems such as AT and 286 class machines lack processing power and storage capacity required for client/server environments.
- **All data is relational:** Many legacy systems use file-based structures, and newer systems may use object-oriented databases that are not purely relational.
- **Development time is always shorter:** Although development tools may reduce time, poor architecture and lack of skills can increase development effort.
- **Open systems solve all problems:** Open standards improve interoperability, but effective management and design are critical for success.

Challenges

The challenges arise due to system heterogeneity, maintenance requirements, organizational restructuring, and cost trade-offs, making client/server computing a strategic rather than a simple solution.

Question 2: Illustrate Any Four Challenges in Adopting Client/Server Architecture in Enterprise Environment

1. **Cost Issues:** While initial hardware costs may be lower than mainframes, hidden costs such as training, integration, and maintenance are significant.
2. **Mixed Platforms:** Enterprises operate with multiple operating systems, network protocols, and hardware platforms, increasing complexity.
3. **Maintenance Complexity:** Client/server systems require continuous maintenance, often costing more than traditional host-based systems.
4. **Organizational Restructuring:** Computing power moves closer to users, while control remains with IS departments, requiring new collaboration models.

Question 3: Importance of a Single System Image in Client/Server Computing and How It Is Achieved

Importance of Single System Image

A Single System Image (SSI) refers to presenting a distributed client/server system as a single, unified system to the user. Its importance includes:

- Simplifies system usage and administration.
- Hides system complexity from end users.
- Improves user productivity.
- Enables seamless access to distributed resources.

Achieving Single System Image

SSI is achieved through:

- Use of open system standards for interoperability.
- Middleware that hides heterogeneity.
- Centralized application servers and databases.
- Uniform user interfaces across platforms.

Question 4: Illustrate and Explain Key Components of Modern Client/Server Architecture

Modern client/server architecture consists of the following components:

1. **Client:** Manages the user interface, validates input, processes part of application logic, and sends requests to the server.

2. **Application Server:** Handles business logic, transaction processing, and coordination between client and database.

3. **Database Server:** Stores and manages data, processes SQL queries, enforces integrity constraints, and ensures recovery.

4. **Middleware:** Acts as a bridge between applications and operating systems, providing communication, security, and portability.

5. **Network:** Enables communication between distributed components using open standards such as TCP/IP.

These components together support scalability, interoperability, data integrity, and efficient resource utilization.